Lab Manual 3: Transient Responses of First and Second Order Circuits

3.1 Introduction and learning objectives

The first part of this lab is to provide hands-on experience and deepen understanding of the step-response behaviour of simple series circuits that include **one** energy-storage element (inductor or capacitor). These circuits can be modelled using a first-order differential equation with either a zero or constant forcing function (commonly referred to as zero state or zero input). In the second part of the lab, it is intended to provide further practical experience and familiarity with the step-responses of simple series circuits which contain **two** energy-storage elements.

The circuits in this lab will be driven by a square wave voltage source so that they can be mathematically described by first-order and second-order differential equation with a zero or a constant forcing function. The detailed mathematical analysis of such circuit has been discussed in lectures. You may need to consult the lecture notes to complete the experiment tasks in this laboratory session.

To perform step response analysis and to study the transient behaviour of RC/RL and RLC circuits when subjected to a sudden change in voltage (a step input), a function generator (Figure 3.1a) is used as the source of step voltages. Meanwhile, voltage waveform measurement across circuit elements will be taken using a digital storage oscilloscope (DSO) (Figure 3.1b).



(a) (b)

Figure 3.1 (a) Function generator; (b) Digital storage oscilloscope

3.2 Tools and Components Needed

Item	*Specification*
Resistors	R = 1 kΩ, 3.3 kΩ, 4.7 kΩ, 6.8 kΩ, 68 Ω, 560 Ω, and 820 Ω
Capacitor	C = 0.1 μF
Inductor	L = 100 mH
Connecting Wires	Assorted
Voltage source	Function generator
Measurement	Digital storage oscilloscope (DSO)

3.3 Experimental Work

3.3.1 RC and RL circuit

1. Configure the generator to produce a 50 Hz square wave with an amplitude of 1 V, ensuring the waveform remains at zero during the other half-period, as illustrated in Figure 3.2.

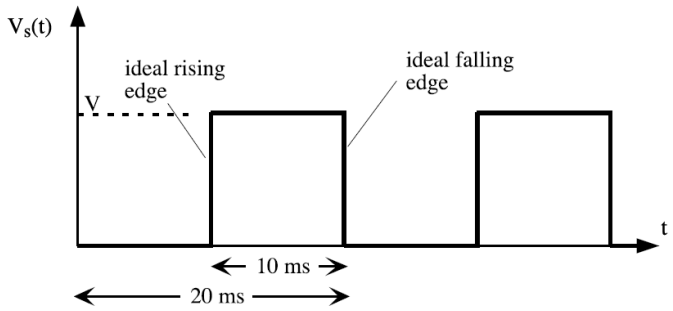


Figure 3.2 A typical ideal square-wave voltage waveform

2. In this series of experiments, the transient behaviour of RC and RL circuits will be examined in Configuration A and B, respectively, as shown in Figure 3.3.

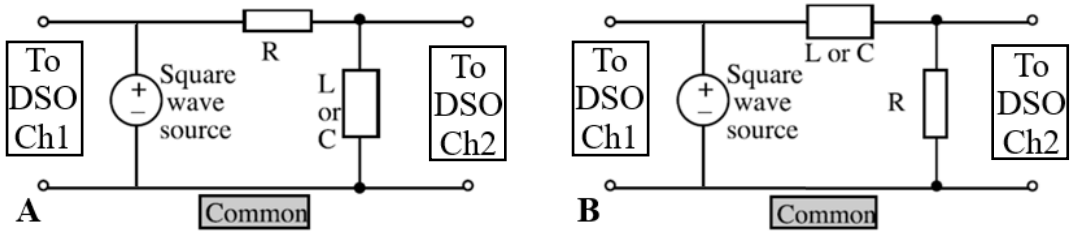


Figure 3.3 (a) Configuration A; (b) Configuration B

3. Connect the elements in Configuration A, starting with RC circuit using 3.3 kΩ resistor and 0.1 μF capacitor.
4. Connect the channel 1 probe of DSO to the terminals of the function generator (square wave source), and channel 2 probe of DSO across the capacitor. **Make sure that the common probes of channel 1 and 2 are connected to the same common terminal of the function generator, and keep the connections simple and secured.**
5. Keep hands off the completed circuit as the DSO can be very sensitive to touches of wires and components during the measurement.
6. Press the Auto Set button at the top right of the DSO to get the initial measurement waveform. Then, adjust the position and scale for horizontal (time) and vertical (amplitude) axis as needed.
7. Examine the train of exponential rising and falling waveforms.
8. Figure 3.4 shows a sketch of a typical exponential waveform arising from a first-order differential equation. Make sure you use the DSO's cursor controls to enable you to measure the time constant as accurately as possible.

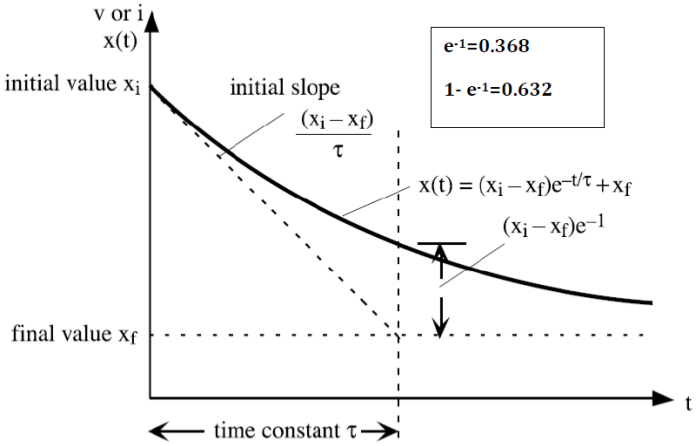


Figure 3.4 Sketch of a typical "first-order" exponential voltage

9. Record down (Take a photo of) the waveforms shown on the DSO's screen, then sketch the **input voltage (Vs)** and **capacitor voltage (Vc) waveforms** to scale on the graph in section 3.4.1 (must show zero-input and zero-state current waveforms flowing in the circuit).

10. Repeat step 3 to 9 for RC circuit with R = 4.7 kΩ and 6.8 kΩ.
11. For RL circuit, connect the same square wave source with R = 68 Ω and L = 100 mH using Configuration A. Measure the **inductor voltage (V_L)** with DSO and record down the waveform.
12. Reconstruct the RL circuit using configuration B and record down the **resistor voltage (V_R)**. Then, sketch the **inductor current (I_L) waveform** to scale based on V_R on the graph in section 3.4.1.
13. Repeat step 12 for R = 560 Ω and 820 Ω.

3.3.2 RLC circuit

1. In the series RLC circuit under study, the "common" terminal of the square wave generator and the "common" terminal of the DSO should be connected together, similar to that of RC and RL circuit.
2. So, it should also be constructed in such a way that the voltage waveform across the element to be observed has a "common" terminal connected to the "common" of the whole circuit.
3. Construct a **series RLC** circuit using 560 Ω, 100 mH, and 0.1 uF. Measure the **capacitor voltage (V_C)** waveform using DSO. Adjust the position and scale for horizontal (time) and vertical (amplitude) axis of the DSO as needed, in order to "zoom-in" the oscillation as shown in Figure 3.5. record down the waveform, then sketch it on the graph in section 3.4.2.

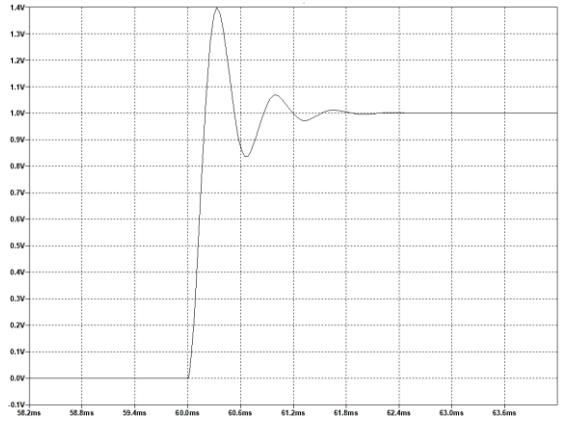


Figure 3.5 Oscillation on second-order transient

4. Use the curser function to measure the period T of the oscillation. (Tip: measure the ΔT from one peak to the next peak or zero-to-zero as shown in Figure 3.6)

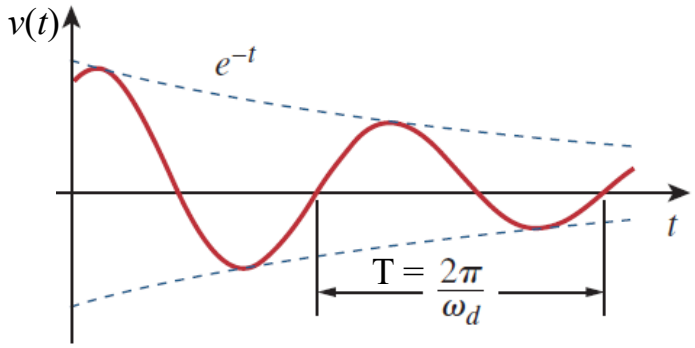


Figure 3.6 Measuring period T of damping oscillation

5. Calculate ω_d based on T, ω_d is the actual oscillation frequency when damping is present.
6. Slowly increase the resistance to R = 1 kΩ, then 2 kΩ, and lastly 3.3 kΩ. Record the changes in the V_C waveform and sketch it on the graph in section 3.4.2.
7. Lastly, remove R from the circuit, where "zero" resistance will cause the damping ratio, $\zeta = 0$. Then, ω_0 is the limiting value of ω_d. Measure T and calculate ω_0 (undamped natural frequency) based on T (Figure 3.7).

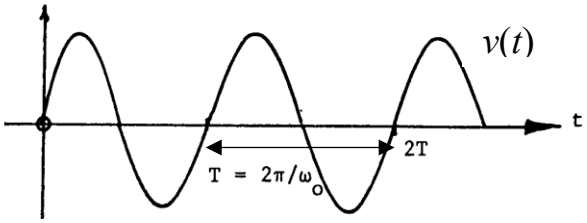
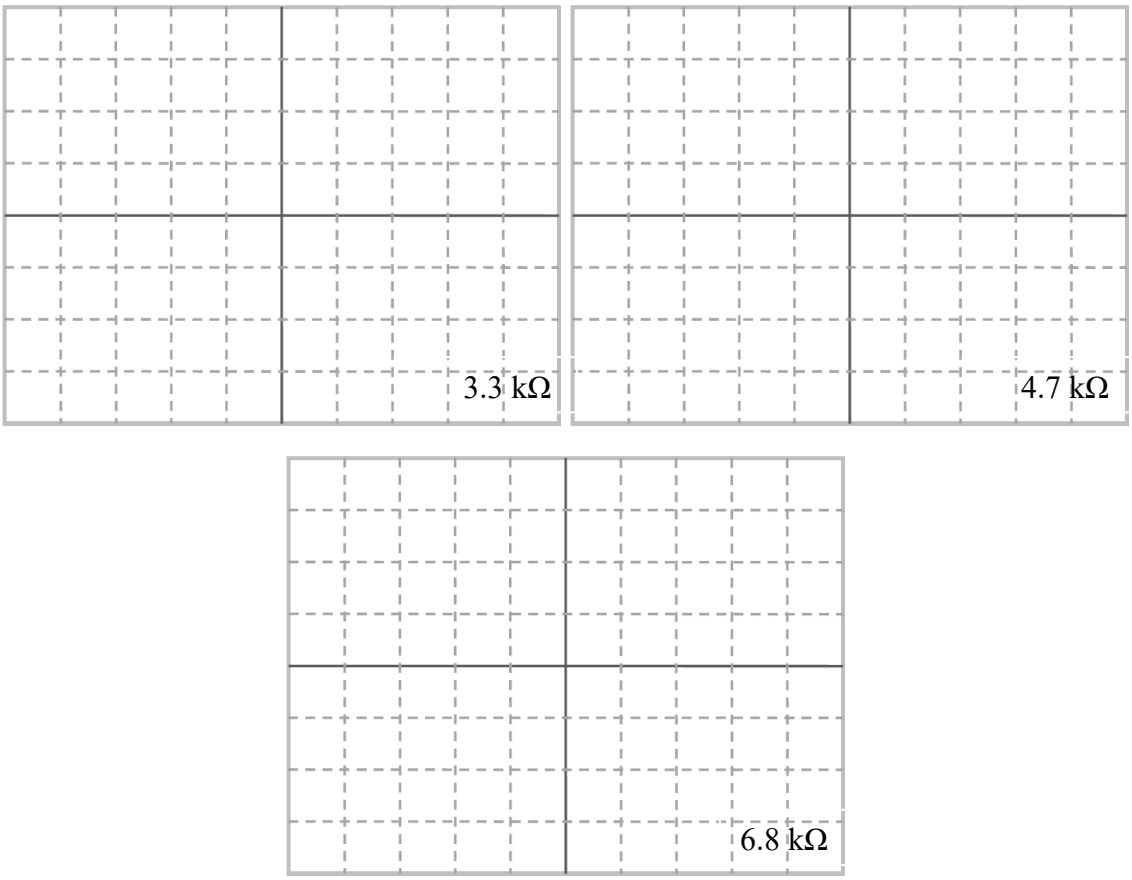


Figure 3.7 Measuring period T of undamped natural oscillation

3.4 Results and Discussion

3.4.1 RC and RL circuit

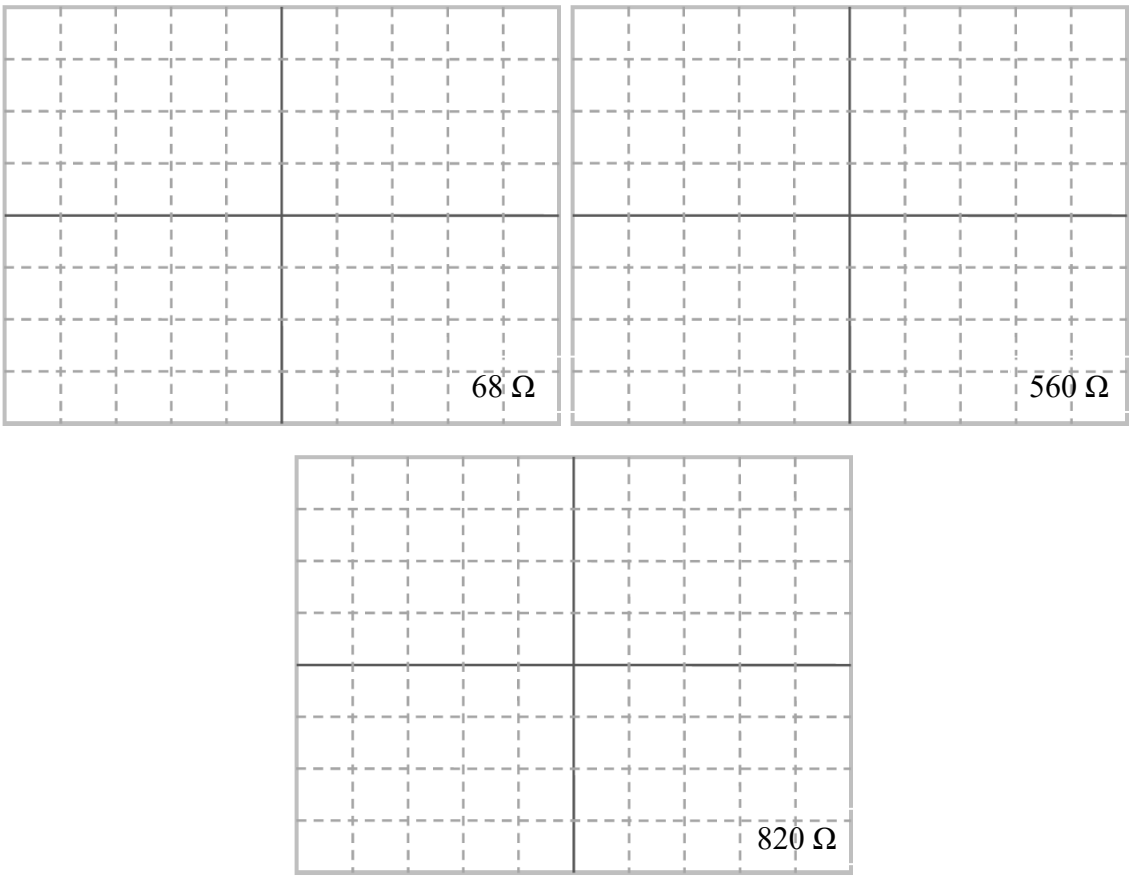
1. Explain why the capacitor voltage V_C in an RC circuit is continuous and cannot change instantaneously. Relate your explanation to the physical properties of a capacitor.
2. On the provided graph below, sketch the source voltage V_S and capacitor voltage V_C to scale. Include both zero-input and zero-state current waveforms. Calculate the time constant τ for each R-C pair using theoretical formulas and experimental data. Clearly mark τ on each curve.



Graph 1: RC circuit's input voltage (V_S) and capacitor voltage (V_C) waveforms

3. Describe the shape of the inductor voltage V_L in the RL circuit and explain why it behaves that way based on the inductor's properties.

4. Sketch the inductor current I_L to scale on the graph below. From the observed waveform, estimate the time constant τ and its uncertainty. Compare this with the theoretical value and mark τ on the graph.

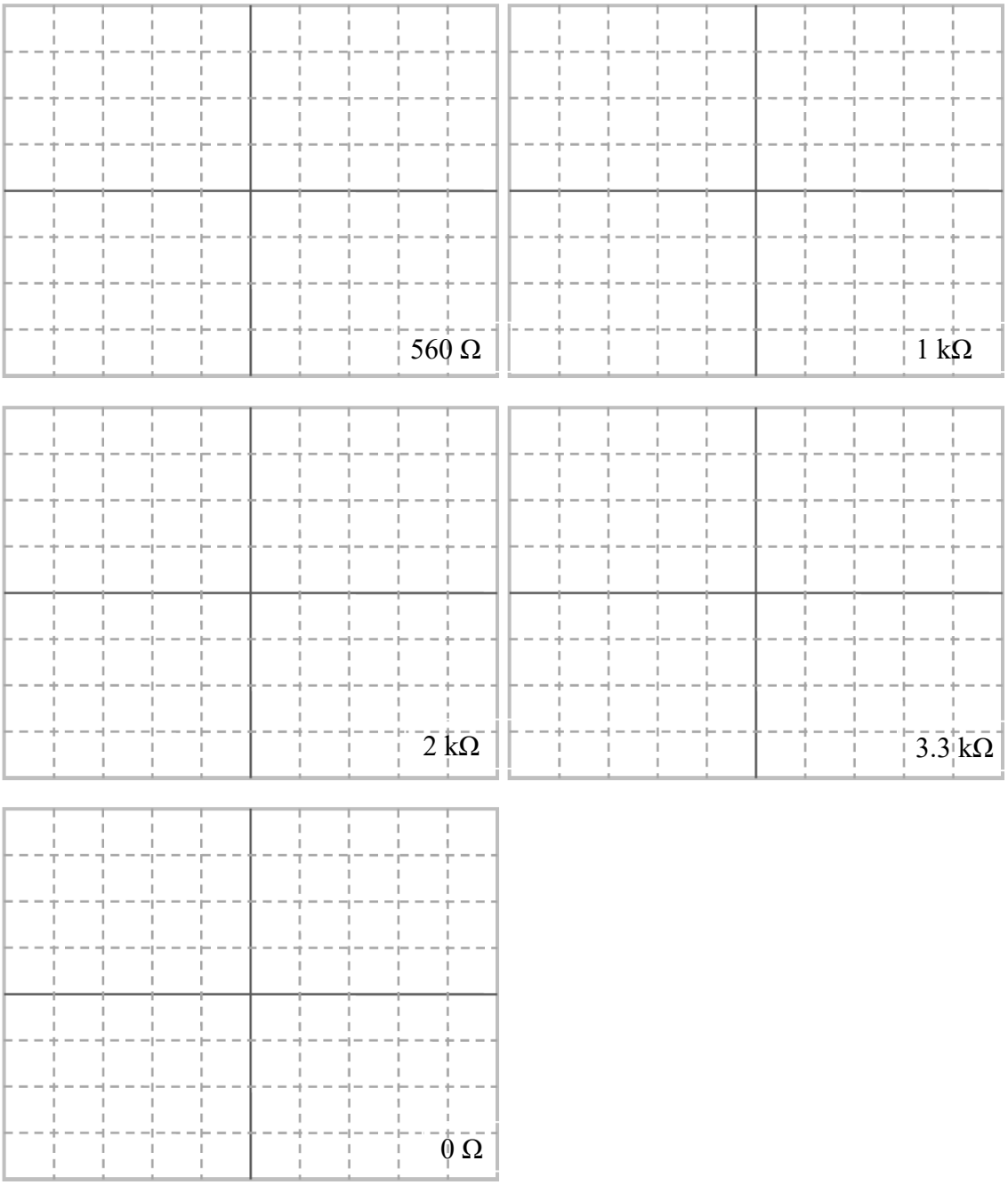


Graph 2: RL circuit's inductor current (I_L) waveforms

3.4.2 RLC circuit

1. Since the total resistance cannot be zero, we approximate ω_0 by minimizing R. List all sources of resistance in the RLC circuit (including components and wiring).
2. Calculate ω_0 from the measured oscillation period T and compare it with the theoretical $\omega_0 = 1/\sqrt{(LC)}$. Discuss any differences and possible reasons.
3. Compute the damped angular frequency ω_d from the observed waveform and compare it with the theoretical value $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$. Explain any discrepancies.

4. For each resistance value, classify the damping as underdamped, critically damped, or overdamped based on α and ω_0.
5. Identify the resistance at which oscillations disappear (critical damping). Compare this experimental value with the theoretical critical resistance.



Graph 3: RLC circuit's capacitor voltage (V_C) waveforms

Report Marking Rubric

Criteria	Poor (1 mark)	Fair (2 marks)	Good (3 marks)	Very Good (4 marks)	Excellent (5 marks)
1. Introduction & Objectives (10%)	Vague or missing; no clear objectives	Basic overview; objectives unclear or incomplete	Adequate summary; some objectives stated	Clear introduction; most objectives covered	Comprehensive intro; all objectives clearly stated and contextualized
2. Methodology: Circuit Assembly, Safety & Measurement Technique (20%)	Circuit not assembled correctly; safety rules ignored; incorrect use of multimeter	Major errors in assembly; minimal safety adherence; some correct measurements	Circuit mostly correct; some safety steps followed; mostly correct technique	Circuit correctly assembled; most safety rules followed; reliable data with few errors	Circuit perfectly assembled; all safety protocols strictly followed; flawless technique and consistent measurements
3. Results: Data Recording & Tables (20%)	Tables missing or incomplete; data unclear	Tables present but poorly formatted or inaccurate	Tables mostly complete; some formatting or data issues	Well-organized tables; mostly accurate data	Clear, complete, and accurate tables with proper formatting
4. Results: Theoretical Calculations, Rule Application, Equivalent Circuit Construction (20%)	No calculations or major errors; no working shown; rules not applied; incorrect or missing equivalents	Attempted calculations; many errors; incomplete working; unclear rule use; unclear circuit construction	Correct approach; some errors in calculation or logic; basic rule application or equivalent circuits	Mostly accurate calculations; clear working steps; rules applied with minor errors; good circuit construction	Fully accurate calculations; all steps clearly shown and justified; rules applied accurately with clear circuit diagrams; equivalents constructed correctly
5. Discussions: Comparison, Discussion & Interpretation of Results (20%)	No comparison or discussion; lacks understanding	Minimal comparison; limited insight	Some comparison; basic discussion of discrepancies	Good comparison; thoughtful discussion of results	Thorough comparison; deep, critical analysis and excellent interpretation of

					discrepancies and implications
6. Conclusion, Report Structure & Presentation (10%)	No sensible conclusion. Disorganized; poor formatting; hard to follow	A conclusion is not supported by any experimental evidence. Some structure; inconsistent formatting	A conclusion is only supported by some experimental evidence. Organized; readable with minor formatting issues	A conclusion is mostly supported by experimental evidence. Well-structured; clear formatting and flow	The aim of the experiment has been achieved with evidence supported conclusion. Professionally presented; logical flow, excellent formatting and clarity

Note to students: Please attach this appendix together with the submission of coursework.